



GAME MANUAL*

1st Edition - August 2026

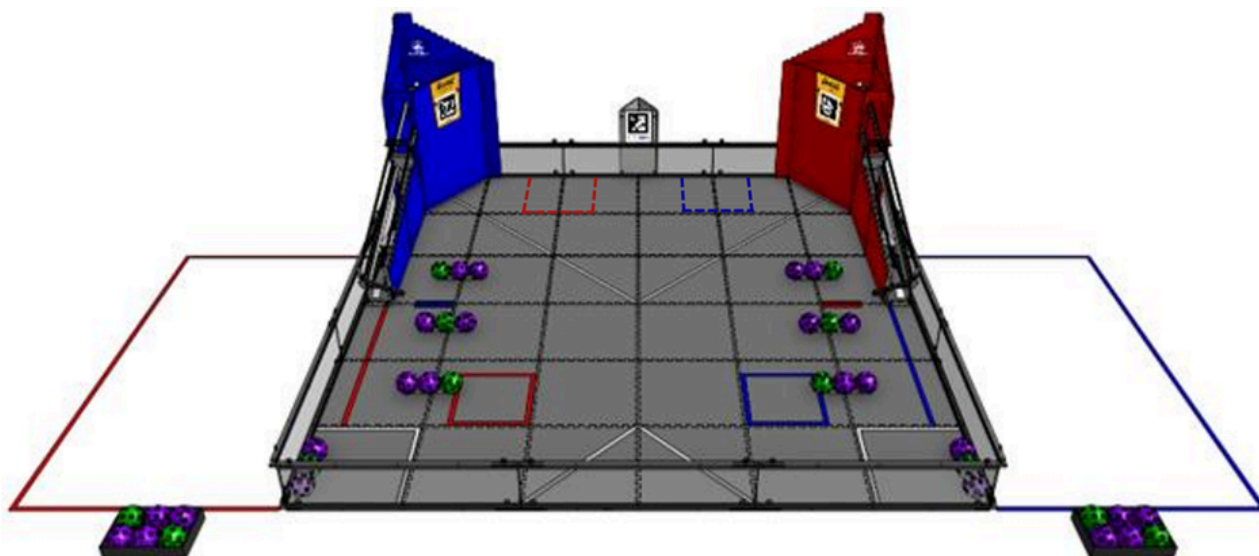
*This manual only contains changes in addition to the last release of the FIRST Tech Challenge DECODE™ 2025–2026 Competition Manual, designed for the SPARKTECH Endurance Cup (SEC) 2026. Any unspecified aspect of the competition is considered identical to the DECODE Competition Manual. For any additional information, consult the Competition Manual.

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CONTENTS	PAGE NUMBER
8. Game Overview.....	3
9. ARENA.....	4
9.3 Areas, Zones & Markings.....	4
10. Game Details.....	5
10.1 MATCH Overview.....	5
10.2 DRIVE TEAM.....	5
10.4 MATCH Periods.....	6
10.5 Scoring.....	7
10.5.4 Point Values.....	7
10.6 Violations.....	8
10.6.2 YELLOW and RED CARD application.....	8
11. Game Rules (G).....	8
11.4 In-MATCH.....	8
11.4.2 TELEOP.....	8
11.4.3 SCORING ELEMENTS.....	9
11.4.4 ROBOT.....	9
11.4.5 Opponent Interaction.....	9
11.4.6 Human.....	10
16. Glossary.....	12
Contact Information.....	12

8. Game Overview

Figure 8-1: FIELD for SEC



In the SEC variation of the FTC DECODE season, ALLIANCES of 2 teams each compete in 12 minute MATCHES by scoring ARTIFACTS in their GOAL, building PATTERNS, and parking in their BASE ZONE in multiple set periods throughout the MATCH.

During the first minute of the MATCH, the ROBOTS operate autonomously. ROBOTS can use sensors to decode the randomized MATCH MOTIF. ROBOTS can earn points by scoring ARTIFACTS in their GOAL and building a PATTERN on their RAMP based on the MOTIF. ROBOTS also earn points for moving off the LAUNCH LINE.

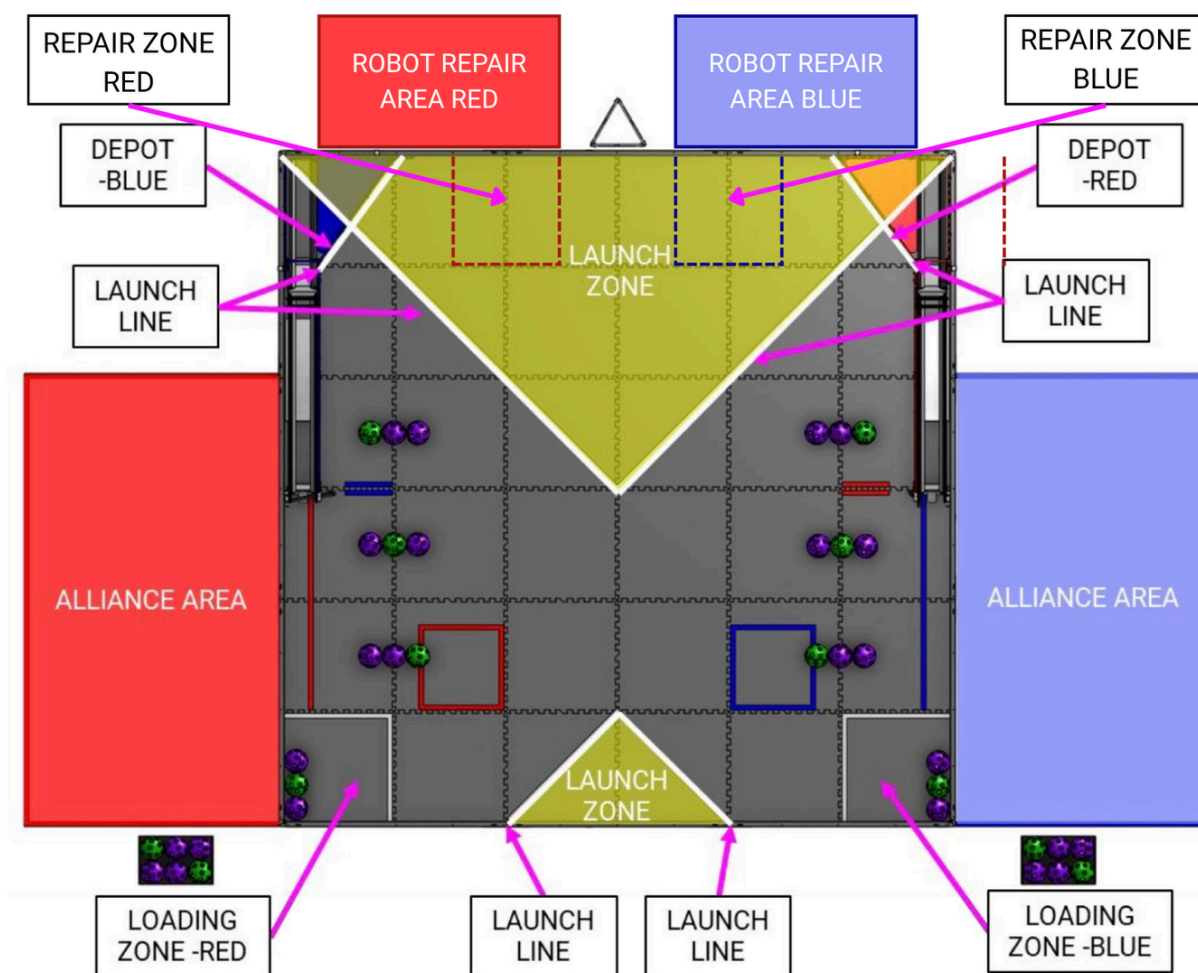
During the remaining 11 minutes, human DRIVERS take control of the ROBOTS. During this MATCH period, ROBOTS have to balance scoring and sorting ARTIFACTS, rushing to their BASE in the 5 PARKING periods.

Whenever needed, the DRIVE TEAM can lift the ROBOTS from their REPAIR ZONES in order to switch batteries and perform maintenance, alternating this aspect with ARTIFACT scoring.

9. ARENA

9.3 Areas, Zones & Markings

Figure 9-1: ALLIANCE AREA, LAUNCH LINES, LAUNCH ZONE and REPAIR ZONE



In addition to the structure presented in the DECODE Competition Manual, the field contains one REPAIR ZONE per ALLIANCE: approximately 24 in. wide by 24 in. deep, infinitely tall volume bounded by intermittent ALLIANCE-specific tape and the adjoining FIELD perimeter situated on the OBELISK side of the FIELD. The REPAIR ZONE is an ALLIANCE-specific zone belonging to the ALLIANCE with the adjacent ALLIANCE AREA (near the opposing ALLIANCE GOAL). While the ROBOT is fully inside its ALLIANCE REPAIR ZONE, the DRIVE TEAM may power off the ROBOT using the power switch, remove it from the FIELD, perform repairs or adjustments in the ROBOT REPAIR AREA, and return the ROBOT to the REPAIR ZONE at any time during the TELEOP period. The ROBOT may be powered back on and re-enter the FIELD only from its ALLIANCE'S REPAIR ZONE.

10. Game Details

10.1 MATCH Overview

MATCHES run on a 15 to 20 minute cycle time per FIELD, which consists of pre-MATCH setup, a minute long AUTO period, a 15-second transition period between AUTO and TELEOP, and an 11-minute TELEOP period consisting of 5 cycles of driving, sorting and parking, followed by the post-MATCH reset. During the MATCH, ROBOTS collect ARTIFACTS and score them into their GOAL to CLASSIFY and create the randomly selected MOTIF. ROBOTS can then open their GATE to continue CLASSIFYING additional ARTIFACTS. ARTIFACTS which do not drop into the RAMP will count as OVERFLOW. ROBOTS conclude the MATCH by returning to their BASE.

10.2 DRIVE TEAM

A DRIVE TEAM is a set of up to 5 people from the same FIRST Tech Challenge team responsible for team performance for a specific MATCH. There are 4 specific roles on a DRIVE TEAM which ALLIANCES can use to assist ROBOTS, and no more than 1 member of the DRIVE TEAM is allowed to be a non-STUDENT.

A DRIVE TEAM may contain, at most:

Role	Description	MAX./ DRIVE TEAM
DRIVE COACH	a guide or advisor	1
DRIVER	an operator and controller of the ROBOT	2
HUMAN PLAYER	a SCORING ELEMENT manager	1
MECHANIC	a student that contributes to the changes on the ROBOT during its removal from the field	1

Note: an ALLIANCE may only contain one HUMAN PLAYER

10.4 MATCH Periods

The first period of each MATCH is one minute long and called the Autonomous Period (AUTO). During AUTO, ROBOTS operate without any DRIVER control or input. There is a 15-second delay between AUTO and TELEOP.

The second period of each MATCH is 10 minutes and 50 seconds (10:50) long and called the teleoperated period (TELEOP). During TELEOP, DRIVERS remotely operate ROBOTS to score points. The TELEOP contains 5 CYCLES of SCORING, each CYCLE is structured in the following way:

- 1 minute and 50 seconds (1:50) period of typical SCORING
- 10 seconds (0:10) of PARKING period (in which the ENDGAME rules apply)
- 10 seconds (0:10) of PARKING BUZZER (in which ROBOTS must be stationary, used for REFEREE scoring purposes)

During each PARKING BUZZER, as well as in the AUTO TRANSITION, BASE and PATTERN points are awarded following the standard Competition Manual rules.

MATCH structure:

Event	Time
AUTO	0:00-1:00
TRANSITION	15 seconds
SCORING period	1:00-2:50
PARKING period	2:50-3:00
PARKING BUZZER	3:00-3:10
SCORING period	3:10-5:00
PARKING period	5:00-5:10
PARKING BUZZER	5:10-5:20
SCORING period	5:20-7:10
PARKING period	7:10-7:20
PARKING BUZZER	7:20-7:30
SCORING period	7:30-9:20
PARKING period	9:20-9:30
PARKING BUZZER	9:30-9:40
SCORING period	9:40-11:30
ENDGAME	11:30-11:40
MATCH END BUZZER	11:40-11:50

10.5 Scoring

SEC MATCHES are organized, played and displayed on a custom scoring interface developed by team SPARKTECH #24345.

10.5.4 Point Values

MATCH Points

For MATCH points, the structure remains identical to the DECODE Competition Manual rules, with the exception of the AUTO LEAVE scoring being changed to 5 points.

Ranking Points

The QUALIFICATION system is modified for the SEC 2026. Each ALLIANCE can now earn up to 10 RANKING POINTS in the following way:

		MATCH points		RANKING
		AUTO	TELEOP	POINTS
LEAVE		5		
ARTIFACT	CLASSIFIED	3	3	
	OVERFLOW	1	1	
	DEPOT		1	
PATTERN	ARTIFACT matches MOTIF	2	2	
BASE	Partially returned to BASE		5	
	Fully returned to BASE		10	
	Additional Bonus: 2 ROBOTS fully returned to BASE		10	
MOVEMENT RP 1 - Combined LEAVE + BASE points earned at or above first threshold				1
MOVEMENT RP 2 - Combined LEAVE + BASE points earned at or above second threshold				1
GOAL RP 1 - The number of ARTIFACTS scored through the SQUARE at or above first threshold				1
GOAL RP 2 - The number of ARTIFACTS scored through the SQUARE at or above second threshold				1
PATTERN RP 1 - PATTERN points earned at or above first threshold				1
PATTERN RP 2 - PATTERN points earned at or above second threshold				1
WIN	Completing a MATCH with more MATCH points than your opponent			4
TIE	Completing a MATCH with the same MATCH points as your opponent			2

RANKING POINT THRESHOLDS:

RP Type	Threshold
MOVEMENT RP 1	60 pts
MOVEMENT RP 2	85 pts
GOAL RP 1	210 ARTIFACTS
GOAL RP 2	300 ARTIFACTS
PATTERN RP 1	50 pts
PATTERN RP 2	72 pts

10.6 Violations

10.6.2 YELLOW and RED CARD application

During an SEC MATCH, YELLOW and RED CARDS are applied by any REFEREE during the match instantly after any violation. If a TEAM / ALLIANCE receives a RED CARD that is confirmed by the Head REFEREE, the MATCH is immediately stopped, and the opposing ALLIANCE is awarded all RANKING POINTS for that MATCH.

11. Game Rules (G)

Adding and removing ROBOTS from the FIELD can be a risk to safety. This is why, in order to keep SEC a safe, yet competitive competition, the violations for interactions in the REPAIR ZONE are severe. Keep each rule in mind and act cautiously in the REPAIR ZONE.

11.4 In-MATCH

11.4.2 TELEOP

- G435** ***ROBOTS are motionless during any of the PARKING BUZZERS.** Any powered movement of the ROBOT or any of its MECHANISMS is not allowed during the PARKING BUZZERS in the TELEOP period.
Violation: MAJOR FOUL. MAJOR FOUL if ROBOT LAUNCHES an ARTIFACT such that it enters the open top of the GOAL or if ROBOT contacts a GATE during a PARKING BUZZER.

11.4.3 SCORING ELEMENTS

G408 ***No more than 3 at a time.** A ROBOT may not simultaneously CONTROL more than 3 ARTIFACTS.

Violation: MINOR FOUL per SCORING ELEMENT over the limit. YELLOW CARD if excessive.

Examples of interaction with a SCORING ELEMENT that are not “CONTROL” include, but are not limited to:

- A. “bulldozing” (inadvertent contact with a SCORING ELEMENT while in the path of the ROBOT moving about the FIELD)
- B. “deflecting” (being hit by a SCORING ELEMENT that bounces into or off a ROBOT)
- C. inadvertent contact with a SCORING ELEMENT while attempting to acquire a SCORING ELEMENT from the LOADING ZONE.
- D. SCORING ELEMENTS that have been LAUNCHED by a ROBOT that are no longer in contact with the ROBOT.

It is important to design your ROBOT so that it is impossible to inadvertently or unintentionally CONTROL more than the limit.

Excessive violations of CONTROL limits include, but are not limited to, frequent (i.e., 3 or more times in a MATCH), simultaneous CONTROL of 5 or more ARTIFACTS.

Strategic violations of G408 (e.g., a strategy to CONTROL and LAUNCH 4 ARTIFACTS) will be considered egregious behavior under G211.

11.4.4 ROBOT

G418 ***ROBOTS may not meddle with ARTIFACTS on RAMPS.** A ROBOT may not:

- A. remove an ARTIFACT from their own RAMP except by operating the GATE, or
- B. remove an ARTIFACT from the opponent’s RAMP by any means

Violation: MAJOR FOUL per ARTIFACT, and the ALLIANCE is ineligible for the PATTERN RP if G418.A, or the opposing ALLIANCE is awarded the PATTERN RP if G418.B.

11.4.5 Opponent Interaction

G427 ***BASE ZONE protection.** During any of the PARKING periods, a ROBOT may not contact, directly or transitively through a SCORING ELEMENT, an opponent ROBOT while either ROBOT is in the opponent’s BASE ZONE, regardless of who initiates contact.

Violation: MAJOR FOUL and opponent ROBOT is awarded fully returned to BASE points.

It is worth noting that one violation of G427 does not guarantee any of the MOVEMENT RANKING POINTS to the opposing ALLIANCE seeing as the THRESHOLDS for any MATCH is higher. However, a violation does offer a significant contribution to obtaining the RANKING POINTS. Keep in mind how close your ALLIANCE is to completing the MOVEMENT RPs during any MATCH.

G436 **Keep your distance from a ROBOT that's being removed / placed.* A ROBOT may not directly or transitively contact a ROBOT that is being removed from or placed onto the FIELD.

Violation: MAJOR FOUL and YELLOW CARD.

For G436, the violation is applied:

A. To the ACTIVE ROBOT, if G439 isn't also applied (the ACTIVE ROBOT was outside of the REPAIR ZONE when the DRIVE TEAM member placed / removed the INACTIVE ROBOT)

B. To the DRIVE TEAM member, if G439 is also applied (the ACTIVE ROBOT was inside of the REPAIR ZONE when the DRIVE TEAM member placed / removed the INACTIVE ROBOT)

G437 **During TELEOP, a ROBOT may not enter or remain in the opposing ALLIANCE'S REPAIR ZONE for longer than MOMENTARY (~3 seconds).*

Violation: MAJOR FOUL; YELLOW CARD if strategic or repeated. YELLOW CARD if an INACTIVE or DISABLED ROBOT obstructs an opponent's REPAIR ZONE.

If a ROBOT loses functionality or connection in the opposing ALLIANCE'S REPAIR ZONE, REFEREES may require the DRIVE TEAM to drive the ROBOT away from the ZONE. If the action is not possible, the ROBOT will become DISABLED and will be treated as obstructing the opponent's REPAIR ZONE.

In the case of obstruction, the ALLIANCE with the obstructed REPAIR ZONE should make a good faith effort to safely place or remove an INACTIVE ROBOT as close as possible to the REPAIR ZONE. Any opponent interaction violations in this case will apply as if the INACTIVE ROBOT was fully inside of the REPAIR ZONE.

11.4.6 Human

G438 **Don't use the REPAIR ZONE to your advantage.* A DRIVE TEAM member may only directly contact their ALLIANCE'S ROBOT situated in the REPAIR ZONE with the intent to power it off from the power switch and remove it from the FIELD.

Violation: MAJOR FOUL.

A DRIVE TEAM member may not use the REPAIR ZONE in order to facilitate any strategic advantage, such as:

- A. Removing an ARTIFACT from the ROBOT
- B. Plugging any cable in or out of the ROBOT
- C. Any minor or major change to a ROBOT'S mechanism

The intent of this rule is to ensure that all ROBOT modifications are made outside of the FIELD.

G439 *No placing / removing ROBOTS from a REPAIR ZONE while another ROBOT is present. A DRIVE TEAM member may not place or remove a ROBOT into the REPAIR ZONE while another ACTIVE ROBOT is in that ZONE (even the ALLIANCE PARTNER ACTIVE ROBOT)
Violation: MINOR FOUL, MAJOR FOUL and YELLOW CARD if this causes a contact between the two ROBOTS.

G440 *Remove ROBOTS safely. A DRIVE TEAM member may not remove a ROBOT from the FIELD while an OPMODE is running.
Violation: MAJOR FOUL and VERBAL WARNING, YELLOW CARD if repeated.

G441 *Don't remove loaded ROBOTS. A DRIVE TEAM member may not remove a ROBOT from the FIELD while it has ARTIFACTS in its possession.
Violation: MINOR FOUL and an additional MINOR FOUL for every 3 seconds in which the situation is not corrected.

This rule's intent is to ensure that ARTIFACTS remain in play even when a ROBOT is out of play.

G442 *Only pick up ROBOTS fully in the REPAIR ZONE. A DRIVE TEAM member may not contact a ROBOT that isn't fully inside of the REPAIR ZONE.
Violation: MAJOR FOUL, YELLOW CARD if the ROBOT isn't fully OR partially inside of the REPAIR ZONE

16. GLOSSARY

MECHANIC - DRIVE TEAM member added to help with maintenance outside of the FIELD

SCORING PERIOD - part of the TELEOP period in which all usual TELEOP rules are applied. There are 5 SCORING PERIODS in each SEC MATCH.

PARKING PERIOD - part of the TELEOP period in which all usual ENDGAME rules are applied. There are 5 PARKING PERIODS in each SEC MATCH, the final one being the ENDGAME PERIOD.

PARKING BUZZER - part of the TELEOP period in which all ROBOTS must be stationary while BASE and PATTERN points are evaluated. There are 5 PARKING BUZZERS in each SEC MATCH, the final one being the MATCH END BUZZER

CYCLE - part of the TELEOP period which consists of 1 SCORING PERIOD, 1 PARKING PERIOD and 1 PARKING BUZZER. There are 5 CYCLES in each SEC TELEOP.

ACTIVE ROBOT - a ROBOT with a running OPMODE.

Contact Information

For more information about the competition rules, venue and registration, check out the @sparktech24345 Instagram account and feel free to contact us!

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